# End-Semester Examination - 2024 (Autumn)
## MATUGBS01
## (Engineering Mathematics – I)
### Semester – I (B. Tech in CEN, MEN, ECE, EEN, CSE/ BCA)

**Full Marks – 80** **Time – 3 Hours**

**Notations and symbols have their usual meaning.**

**Use separate answer sheets for Group – A and Group – B**

**Group–A (40 marks)**

**1.** Answer all the following questions: $1 \times 5 = 5$

(a) Define convergent sequence in Real number.

(b) Write the definition of oscillatory series.

(c) Define local maxima and local minima of a function.

(d) Write the statement of L'Hospital theorem.

(e) State the relation between the Gamma and Beta function.

**2.** Answer all the following questions: $1 \times 5 = 5$

(a) The value of the $\lim\limits_{x \to 1} \dfrac{1-x^2}{x^2+x-2}$ is

(1) $-2$
(2) $-\frac{2}{3}$
(3) $1$
(4) $2$

(b) The value of the $\beta(4,7) =$

(1) $\frac{2}{99}$
(2) $\frac{1}{99}$
(3) $\frac{1}{33}$
(4) $-\frac{4}{99}$

$\left(\frac{1}{840}\right)$

(c) Find the value of c guaranteed by the Rolle's theorem of $f(x) = 3x^3 - 4x$ in $[-1, 0]$

(1) $\frac{2}{3}$
(2) $\frac{4}{9}$
(3) $-\frac{2}{3}$
(4) $-\frac{4}{9}$

(d) If a series $\sum x_n$ is convergent then which of the following always occurs

(1) $\lim\limits_{n \to \infty} x_n = 1$
(2) $\lim\limits_{n \to \infty} x_n = \infty$
(3) $\lim\limits_{n \to \infty} x_n = 0$
(4) $\lim\limits_{n \to \infty} x_n \neq 0$

(e) Which of the following is not correct

(1) $\lim\limits_{n\to\infty}\frac{1}{n}$ is convergent

(2) $\lim\limits_{n\to\infty}\frac{1}{n}$ is divergent

(3) $\lim\limits_{n\to\infty}\frac{1}{n^2}$ is convergent

(4) $\lim\limits_{n\to\infty}\frac{1}{\sqrt{n}}$ is divergent

**3.** Answer any **SIX** of the following questions: $5\times 6=30$

(a) Show that the below sequence is convergent $4+1$

$\sqrt{3}, \sqrt{3+\sqrt{3}}, \sqrt{3+\sqrt{3+\sqrt{3}}}, \sqrt{3+\sqrt{3+\sqrt{3+\sqrt{3}}}}, \ldots$ Find out where it converges.

(b) Test the convergence of the following series by Cauchy's root test. 5

$$\sum_{n=1}^{\infty}\left(\frac{n}{n+1}\right)^{n^2}$$

(c) Test the convergence of the series 5

$$\frac{2}{1\cdot 3}+\frac{3}{3\cdot 5}+\frac{4}{5\cdot 7}+\cdots$$

(d) Find the value of $x$ for which the function $f(x)=x^3-6x^2+9x+7$ has maximum and minimum. Find the global maximum of and minimum values in $[0,6]$. 5

(e) (i) Show that $\lim\limits_{x\to 0}\cos\left(\frac{1}{x}\right)$ does not exist. $4+1$

(ii) Define the radius of convergence.

(f) Verify the Rolle's theorem of the following function $f(x)=2x^3+x^2-4x-2$, in the interval $-\sqrt{2}\le x\le\sqrt{2}$. 5

(g) Show that the function $f(x)$ is continuous at the points $6, 0,$ and $-2$, where, $f(x)=|x|-|x+2|+|x-6|$. 5

(h) Evaluate by L' Hospital rule, $\lim\limits_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\log x}\right)$. 5

(i) If $m$ is a positive integers, then $\beta(m,n)=\frac{(m-1)!}{n(n+1)(n+2)\cdots(n+m-1)}$. 5

## Group - B (40 marks)

**1.** Answer all the following questions: $1 \times 5 = 5$

(a) The line $\frac{x-1}{2} = \frac{y-1}{1} = \frac{z-0}{-1}$ is perpendicular to the line

~~(1)~~ $\frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$

(2) $\frac{x-2}{1} = \frac{y-5}{1} = \frac{z-0}{1}$

(3) $\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-7}{1}$

(4) $\frac{x-2}{0} = \frac{y-4}{1} = \frac{z-0}{-1}$

(b) The distance between two planes $12x - 4y + 3z + 10 = 0$ and $12x - 4y + 3z + 36 = 0$ is

(1) 1

~~(2)~~ 2

(3) 3

(4) 4

(c) The angle between two planes $x + y + 1 = 0$ and $y + z + 1 = 0$ is

(1) 0

(2) $\frac{\pi}{2}$

~~(3)~~ $\frac{\pi}{3}$

(4) $\frac{\pi}{4}$

(d) If $f(x,y,z) = 3x^2y - y^3z^2$, then grad $f$ at $(1, -2, -1)$ is equal to

(1) $12i + 9j + 16k$

~~(2)~~ $-12i - 9j - 16k$

(3) $i + 9j + 16k$

(4) $i + 9j - 16k$

(e) For which type of curve can we use Green's theorem?

(1) open curve

~~(2)~~ closed curve

(3) both for open and closed

(4) all of above.

**2.** Answer all the following questions: $1 \times 5 = 5$

(a) Convert the point $(7, -7\sqrt{3})$ in to the polar coordinate.

(b) Find out the direction cosine of the straight line $\frac{2x-1}{6} = \frac{2-y}{4} = \frac{2z-3}{24}$.

(c) What do you mean by the curvature of a curve?

(d) What do you mean by the arc length of a curve?

(e) Find out the value of the double integral $\int_0^1 \int_0^1 (x^2)\, dxdy$

**3.** Answer any **SIX** of the following questions: $5 \times 6 = 30$

(a) I) If a line makes an angle $\frac{\pi}{3}$ with $x$-axis, $\frac{\pi}{3}$ with $y$-axis and $\theta \in (0, \pi)$ with $z$-axis, then find out the value of $\theta$. $2\frac{1}{2}$

II) Find out the nature of the quadratic surface $2x^2 + 5y^2 + 3z^2 - 4x + 20y - 6z = 5$ and hence evaluate its volume. $2\frac{1}{2}$

(b) Find out the image of the point $(-3, 8, 4)$ with respect to the plane $6x - 3y - 2z + 1 = 0$. 5

(c) Write down the equation of the line that is perpendicular to both lines given below; hence, find out the distance between the lines given below 5

$$\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-0}{0} \text{ and } \frac{x-1}{0} = \frac{y-1}{1} = \frac{z-2}{1}$$

(d) Find out the equation of the plane passing through the point $(1,1,2)$ and perpendicular to the planes $x-y+z=1$ and $2x-3y+z=1$. 5

(e) Find out the curvature at any point $t$ of the curve 5

$$F(t)=(\cos t,-\sin t,2t).$$

(f) Find out the torsion of the curve 5

$$F(t)=(e^{t},e^{-t},\sqrt{2}t).$$

(g) Evaluate

$$\iint (x+y)\,dxdy$$

over the region enclosed by the circle of radius 2, $y=x$ and the positive $x$ axis. 5

(h) Evaluate

$$\iiint (x+y+z+1)dxdydz$$

over the region defined by $x\geq 0,\ y\geq 0,\ z\geq 0,\ x+y+z\leq 1$. 5

(i) Use Green's theorem to evaluate

$$\oint_C (3xy^2-6y)dx+(y-3xy)dy$$

where $C$ is the boundary of the region $x=0,\ \ y=0,\ \ 2x+3y=12$. 5

(j) Use Divergence theorem to evaluate

$$\iint_S \vec{F}\cdot d\vec{S}$$

where $\vec{F}=\cos(\pi x)\hat{\imath}+2xy^3\hat{\jmath}+(z^2-4x)\hat{k}$ and $S$ is the surface of the box with $-1\leq x\leq 2,\ 0\leq y\leq 1$ and $1\leq z\leq 4$. 5